Year 11 Set 3 · GCSE · grade 4–5 · Lesson 1

Expanding and factorising, including quadratics

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Today I will learn to...

Goals

1. Expand single and double brackets

2. Factorise expressions and quadratics

Do now: retrieval practice

4 mins

1. Simplify 3a + 5b − a + 2b

   2a + 7b

2. Work out −4 × −6

   24

3. Find two numbers that multiply to 12 and add to 7

   3 and 4

4. Expand 5(2x − 3)

   10x − 15

Objective 1

Expand single and double brackets

Key idea

Learn

!

Multiply everything inside the bracket by the term outside. For two brackets, multiply each term in the first bracket by each term in the second (4 multiplications), then collect like terms. Watch the signs: − × − = +.

Objective 1: Expand single and double brackets

I do: example 1

I do

Expand and simplify $3(x + 4) - 2(x - 5)$

1. $3(x + 4) = 3x + 12$
2. $-2(x - 5) = -2x + 10$
3. $3x + 12 - 2x + 10$
4. $= x + 22$

I do: example 2

I do

Expand and simplify $(x + 3)(x + 5)$

1. $x \times x = x^2$
2. $x \times 5 = 5x$ and $3 \times x = 3x$
3. $3 \times 5 = 15$
4. $x^2 + 8x + 15$

I do: example 3

I do

Expand and simplify $(2x - 1)(x + 4)$

1. $2x \times x = 2x^2$
2. $2x \times 4 = 8x$ and $-1 \times x = -x$
3. $-1 \times 4 = -4$
4. $2x^2 + 7x - 4$

Mini whiteboards

We do

1. Expand $4(3x - 2)$

Answer: $12x - 8$

2. Expand and simplify $(x + 6)(x - 2)$

Answer: $x^2 + 4x - 12$

3. Expand and simplify $(x - 4)(x - 6)$

Answer: $x^2 - 10x + 24$

4. Expand and simplify $(3x + 1)(x - 2)$

Answer: $3x^2 - 5x - 2$

You do: choose your challenge

You do

Bronze

Expand $5(x - 4)$

Silver

Expand and simplify $2(3x + 1) + 4(x - 3)$

Gold

Expand and simplify $(x - 5)(x - 3)$

Platinum

Expand and simplify $(2x + 3)(3x - 4)$

You do: answers

Check

Bronze

$5x - 20$

Silver

$6x + 2 + 4x - 12 = 10x - 10$

Gold

$x^2 - 8x + 15$

Platinum

$6x^2 - 8x + 9x - 12 = 6x^2 + x - 12$

If you finish early

Extra

Bronze

Expand $6(2x + 5)$

$12x + 30$

Silver

Expand and simplify $(x + 9)(x - 9)$

$x^2 - 81$

Gold

Expand and simplify $(2x - 3)^2$

$4x^2 - 12x + 9$

Platinum

A square has side $(x + 5)$ cm. A rectangle is $(x + 2)$ cm by $(x + 8)$ cm. Which has the bigger area, and by how much?

Square $x^2 + 10x + 25$, rectangle $x^2 + 10x + 16$: the square, by 9 cm^2

Exam question 1

2 marks

Expand and simplify $(x - 7)(x + 3)$

Space for working

Mark scheme: Three of x^2, $3x$, $-7x$, -21 correct (M1). $x^2 - 4x - 21$ (A1)

Exam question 2

2 marks

Expand and simplify $(3x + 2)^2$

Space for working

Mark scheme: $(3x + 2)(3x + 2)$: three of $9x^2$, $6x$, $6x$, 4 (M1). $9x^2 + 12x + 4$ (A1)

Objective 1 achieved

Expand single and double brackets



Objective 2

Factorise expressions and quadratics

✓ Achieved: Expand single and double brackets

Key idea

Learn

!

Factorising is the reverse of expanding. Take out the highest common factor first. For $x^2 + bx + c$, find two numbers that multiply to c and add to b. A difference of two squares $a^2 - b^2 = (a + b)(a - b)$.

Objective 2: Factorise expressions and quadratics

I do: example 1

I do

Factorise fully $6x^2 + 9x$

1. HCF of 6 and 9 is 3
2. Both terms contain x
3. HCF is 3x
4. $3x(2x + 3)$

I do: example 2

I do

Factorise $x^2 + 7x + 10$

1. Multiply to 10, add to 7
2. Factor pairs of 10: 1 and 10, 2 and 5
3. $2 + 5 = 7$ ✓
4. $(x + 2)(x + 5)$

I do: example 3

I do

Factorise $x^2 - 49$

1. It is a difference of two squares
2. $x^2 = x \times x$ and $49 = 7 \times 7$
3. $a^2 - b^2 = (a + b)(a - b)$
4. $(x + 7)(x - 7)$

Mini whiteboards

We do

1. Factorise $8y - 12$

Answer: $4(2y - 3)$

2. Factorise $x^2 - x - 12$

Answer: $(x - 4)(x + 3)$

3. Factorise $x^2 + 11x + 24$

Answer: $(x + 3)(x + 8)$

4. Factorise $x^2 - 36$

Answer: $(x + 6)(x - 6)$

You do: choose your challenge

You do

Bronze

Factorise $7x + 21$

Silver

Factorise $x^2 + 9x + 20$

Gold

Factorise $x^2 - 3x - 18$

Platinum

Factorise $4x^2 - 25$

You do: answers

Check

Bronze	$7(x + 3)$
Silver	$(x + 4)(x + 5)$
Gold	$(x - 6)(x + 3)$
Platinum	$(2x + 5)(2x - 5)$

If you finish early

Extra

Bronze

Factorise $15x - 10$

$5(3x - 2)$

Silver

Factorise $x^2 - 8x + 15$

$(x - 3)(x - 5)$

Gold

Factorise $x^2 + 4x - 21$

$(x + 7)(x - 3)$

Platinum

Use factorising to work out $53^2 - 47^2$ without a calculator

$(53 + 47)(53 - 47) = 100 \times 6 = 600$

Exam question 1

2 marks

Factorise $x^2 + 2x - 15$

Space for working

Mark scheme: (x + 5)(x − 3) (B2; B1 for one correct bracket)

Exam question 2

2 marks

Factorise fully $10a^2b - 15ab^2$

Space for working

Mark scheme: 5ab(2a − 3b) (B2; B1 for a correct partial factorisation)

Objective 2 achieved

Factorise expressions and quadratics



Real life

Apply

A rectangular garden is $(x + 4)$ m long and $(x + 2)$ m wide. Write an expression for its area.

$(x + 4)(x + 2) = x^2 + 6x + 8$ m²

Extension: stretch yourself

Stretch

1. Expand and simplify $(x + 2)(x + 3)(x - 1)$

$(x^2 + 5x + 6)(x - 1) = x^3 + 4x^2 + x - 6$

2. Factorise $2x^2 + 7x + 3$

$(2x + 1)(x + 3)$

Exit ticket: exam questions

Exit

1. Expand and simplify $(x + 4)(x - 6)$ **(2 marks)**

2. Factorise $x^2 + 8x + 12$ **(1 mark)**

3. Factorise $x^2 - 81$ **(1 mark)**

Exit ticket: mark scheme

Mark it

Q1 $x^2 - 6x + 4x - 24$ (M1) $= x^2 - 2x - 24$ (A1) [2]

Q2 $(x + 2)(x + 6)$ [1]

Q3 $(x + 9)(x - 9)$ [1]

Total: ____ / 4

Today I achieved

- ✔ **Expand single and double brackets**
- ✔ **Factorise expressions and quadratics**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Expand and simplify $4(2x + 3) - 3(x - 1)$ [2]
2. Expand and simplify $(x + 7)(x - 2)$ [2]
3. Factorise fully $12x^2 - 8x$ [2]
4. Factorise $x^2 - 9x + 14$ [2]
5. Factorise $x^2 - 100$ [1]

Extra practice: Maths Genie → Edexcel question bank → Expanding and factorising quadratics